1. What is measure of dispersion? Explain any four measures of dispersion.
2. What is measure of dispersion? Explain variance and coefficient of variance.
3. What is measure of dispersion? Explain mean deviation about mean and standard deviation.
4. The mean of 5 observations is 4·4 and variance is 8·24. It three of the five observation are 1, 2 and 6; find the other two.
5. The mean and standard deviation of a group of 10 observations were found to be 20 and 5 respectively. Later on it was found that the observations were incorrectly taken as 25 instead of 5. Find the corrected mean and standard deviation.
6. The mean and standard deviation of marks of a group of100 students were found to be 60 and 15 respectively. Later on it was found that the scores 45 and 72 were wrongly entered as 40 and 27 respectively. Find the corrected mean and standard deviation.
7. The mean and standard deviation of 30 items are 20 and 3 respectively out of these 30 items, item 22 and 15 are dropped. Find new A.M. and S.D. Calculate A.M. and S.D. if item 22 is replaced by 8 and 15 is replaced by 17.
8. For a group of 200 candidates, the mean and standard deviation of scores were found to be 40 and 15 respectively. Later on it was discovered that the scores 43 and 35 respectively. Find the corrected mean and standard deviation corresponding to the corrected figures.
9. The tax authority collected the following amount of tax from different firms in a particular market. Calculate the quartile deviation and the coefficient of quartile deviation

Amount of taxes (in Thousand)	10	11	12	13	14
Frequency	3	12	18	12	3

10. Find the quartile deviation and coefficient of quartile deviation of the following frequency distribution

Amount of taxes (in Thousand)	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	10	20	30	50	40	30

11. Find out the quartile deviation for the data

Class Interval	0-10	10-20	20-30	30-40	40-50
Frequency	3	5	7	9	4

12. Calculate Mean Deviation from mean for the following data

x	1	2	3	4	5	6	7
f	3	5	8	12	10	7	5

13. Calculate mean, mean deviation from mean and co-efficient of mean deviation.

Marks	0-10	10-20	20-30	30-40	40-50
No. of students	5	8	15	16	6

14. Find the Mean deviation from median for the following data

Class Interval	0-10	10-20	20-30	30-40	40-50
Frequency	3	5	7	9	4

15. Calculated mean deviation and co-efficient of mean deviation from median

No. of tomatoes per plant	0	1	2	3	4	5	6	7	8	9	10
No. of plants	2	5	7	11	18	24	12	8	6	4	3

16. Runs scored in 10 matches of IPL season by two batsmen A and B are tabulated as under

Batsman A	46	34	52	78	65	81	26	46	19	47
Batsman B	59	25	81	47	73	78	42	35	42	10

Decide who is better batsman and who is more consistent.

17. Calculate standard deviation for the following frequency distribution. Decide whether A.M. is good average.

Wages in Rupees earned per day	0-10	10-20	20-30	30-40	40-50	50-60
No. of laborers	5	9	15	12	10	3

18. Fluctuations in the Aggregate of marks obtained by two groups of students are given below. Find out which of the two shows greater variability.

Gr-A	518	519	530	530	544	542	518	550	527	527	531	550	550	529	528
Gr-B	825	830	830	819	814	814	814	844	842	842	826	832	835	840	840

19. An analysis of monthly wages paid to tile workers of two firms A and B belonging to the same industry gives tile following results:

	Firm A	Firm B
Number of workers	500	600
Average monthly wages	Rs. 186	Rs. 175
Variance of distribution of wages	81	100

a) Which firm has a larger wages bill?

b) In which firm is there greater variability in individual wage?